



CONNECTIVE TISSUES

TOPIC OVERVIEW

I. OVERVIEW.....	1
A. CLASSIFICATION.....	2
II. EMBRYONIC GERM LAYERS.....	2
III. EMBRYONIC CONNECTIVE TISSUES.....	2
IV. MESENCHYME.....	2
A. MESENCHYME ECM COMPOSITION.....	2
V. MUCOUS CONNECTIVE TISSUE (WHARTON'S JELLY).....	2
VI. CONNECTIVE TISSUE PROPER.....	3
A. LOOSE CONNECTIVE TISSUE (AREOLAR TISSUE).....	3
B. DENSE (IRREGULAR) CONNECTIVE TISSUE.....	3
C. DENSE (REGULAR) CONNECTIVE TISSUE.....	3
VII. TENDONS.....	4
VIII. LIGAMENTS.....	4
IX. APONEUROSSES.....	4
X. CONNECTIVE TISSUE FIBERS.....	4
A. COLLAGEN FIBERS.....	4
B. RETICULAR FIBERS.....	5
C. ELASTIC FIBERS.....	6
XI. CLINICAL CORRELATION: COLLAGENOPATHIES.....	6
XII. EXTRACELLULAR MATRIX.....	6
XIII. GROUND SUBSTANCE.....	7
E. GLYCOSAMINOGLYCANS (GAGs).....	7
B. PROTEOGLYCANS.....	7
C. MULTIADHESIVE GLYCOPROTEINS.....	7
XIV. CONNECTIVE TISSUE CELLS.....	7
A. FIBROBLASTS.....	8
B. ADIPOCYTES.....	8
C. MACROPHAGES.....	8

D. MAST CELLS.....	8
XV. MONONUCLEAR PHAGOCYTE SYSTEM.....	9
A. ADULT STEM CELLS.....	9
B. PERICYTES (PERIVASCULAR CELLS).....	9
C. WANDERING CELLS.....	9
XVI. ADIPOSE TISSUE.....	10
A. WHITE ADIPOSE TISSUE.....	10
B. BROWN ADIPOSE TISSUE.....	10
C. BEIGE ADIPOSE TISSUE.....	10
XVII. REFERENCES.....	11

I. OVERVIEW

- Unlike epithelial tissue, connective tissue is **more diverse in its structure and arrangement**.
- Connective tissue comprises a diverse group of cells within a tissue-specific **extracellular matrix (ECM)**.
- ECM Components:**
 - Protein fibers:** collagen, elastic, reticular fibers
 - Ground substance:** glycosaminoglycans (GAGs), proteoglycans, glycoproteins
 - Ground substance forms **viscous material** in **basal laminae**.
 - The support structure.
 - They form this vast continuous compartment throughout the body bounded by the basal lamina of various epithelia or by the external laminae of muscle and nerve cells.

- Different connective tissues have different functions based on cell type, fiber arrangement, or composition.
- Key Functions:**
 - Supports and binds together structural elements throughout the body.
 - Functions vary based on cell types, fiber arrangements, and ECM composition.

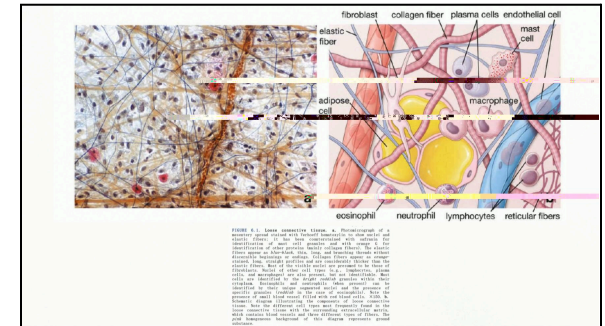


Diagram 1. Loose Connective Tissue

- Shows loose connective tissue.
- Made up of 2 main components:**
 - Cells
 - Extracellular matrix
- Various cells scattered throughout:**
 - Fibroblasts
 - Mast cells
 - Macrophage
- ECM surrounds the cells and is composed of various proteins and fibers.
- Embedded in the ground substance, shown as a pink background material.

- **Connective tissue** is cells plus ECM.

A. CLASSIFICATION

- Mainly based on composition and organization of extracellular components and cells.
 1. **Embryonic connective tissue**
 2. **Connective tissue proper (Loose or dense connective)**
 3. **Specialized connective tissues: Cartilage, bone, adipose, blood, hematopoietic tissue**
- In the past, **elastic and reticular tissues** were separated as different variations, but now they are understood as types of loose or dense irregular connective tissues based on the predominance of fibers or cell types.

II. EMBRYONIC GERM LAYERS

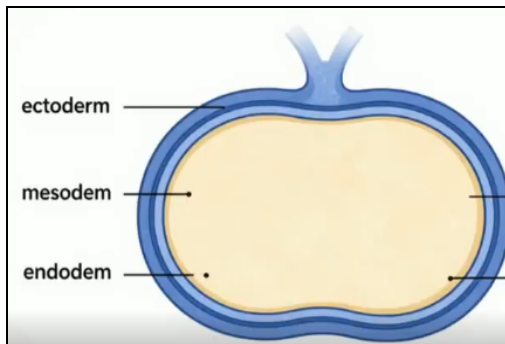


Diagram 2. Embryonic Germ Layers

1. **Ectoderm** — outer layer; nervous system, skin epidermis
2. **Mesoderm** — middle layer; connective tissues, muscle, blood, bone.
3. **Endoderm** — inner layer; digestive tract, respiratory system

III. EMBRYONIC CONNECTIVE TISSUES

- Precursor to various tissues in the adult body.
- Originates from the mesenchyme, which originates from the mesoderm.

Mesoderm (Middle Embryonic Germ Layer)

- Gives rise to almost all connective tissues.
- **Exception:** Head region connective tissue derived from ectoderm via neural crest cells.
- Connective tissue in your nervous system is not derived from the mesoderm.
- Present in the embryo and within the umbilical cord.
- **Two subtypes:**
 - Mesenchyme
 - Mucous connective tissue

IV. MESENCHYME

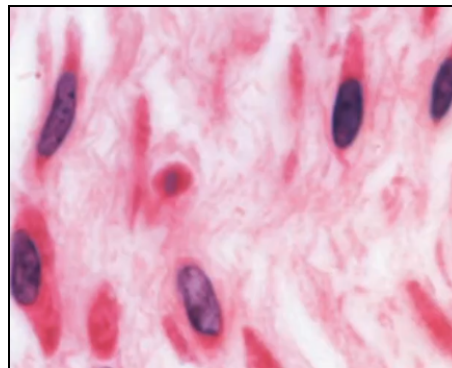


Diagram 3. Mesenchyme

- Primarily found in the **embryo**.
- Came from the proliferation and migration of the **mesoderm and neural crest cells**.
- Not only give rise to the connective tissues of the adult but also **muscle, vascular and urogenital system**, and **serous membranes of the body cavities**.

- Contains **small, spindle-shaped** cells of **relative uniform appearance**.
- Cells extend processes from contact with similar processes of neighbouring cells forming a **3D cellular network**.
- **Gap junctions** are present where these processes make contact.

A. MESENCHYME ECM COMPOSITION

- Viscous ground substance.
- Fine collagen and reticular fibers.
- Fibers are fine and relatively sparse.
- The viscous fibers are consistent with the limited physical stress of the embryo.

V. MUCOUS CONNECTIVE TISSUE (WHARTON'S JELLY)

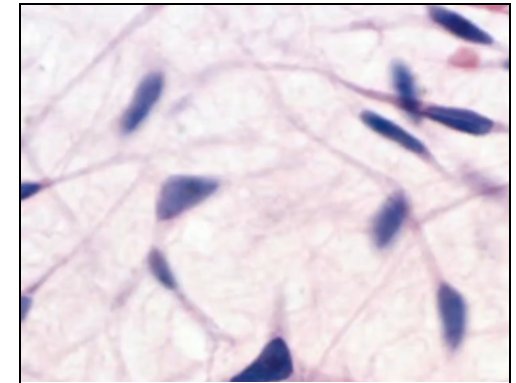


Diagram 4. Mucous Connective Tissue

- Specifically found in the **umbilical cord**.
- Consists of **specialized gelatinous ECM** composed of mainly **hyaluronan**.
- **Ground substance = Wharton's jelly**; occupies large intercellular spaces.
- Spindle-shaped fibroblasts are widely separated by thin wispy collagen fibers.
- Contains mesenchymal stem cells with differentiation potential.

- In routine H&E preparation it occupies large intercellular spaces located in between the wispy collagen fibers.

VI. CONNECTIVE TISSUE PROPER

- **Loose Connective Tissue (Areolar Tissue)**
- **Dense Connective Tissue**
 - Dense **Irregular** Connective Tissue
 - Dense **Regular** Connective Tissue

A. LOOSE CONNECTIVE TISSUE (AREOLAR TISSUE)

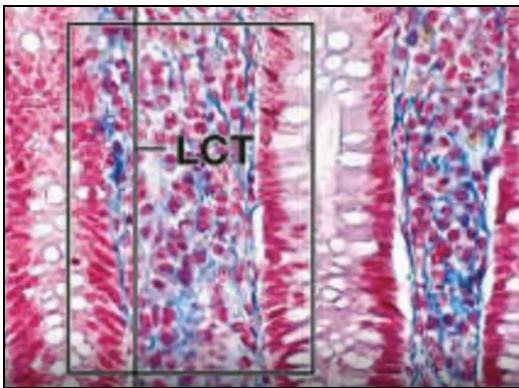


Diagram 5. Loose Connective Tissue

- **Loosely arranged fibers** with abundance of cells of various types
- It's a cellular tissue with thin, and relatively sparse collagen fibers; ground substance is more abundant and occupies more volume than fibers
- **Viscous to gel-like consistency**

Critical Roles:

- **Diffusion of oxygen and nutrients** from small vessels and disconnection tissue
- **Diffusion of carbon dioxide and metabolic wastes** back to the tissue

Location:

- Primarily located **beneath epithelia (Lamina Propria)**; surrounds epithelium of glands and blood vessels
 - Lie in between the epithelia that cover surfaces and line the internal surface of the body

Immune Response:

- Initial site of **pathogenic infiltration** and **immune response**
 - Bacteria that breach the surface are challenged and destroyed by the immune cells
 - Most cells in the connective tissue are transient wandering cells that migrate from blood vessels in response to stimuli
 - Site of inflammatory and immune reactions

B. DENSE (IRREGULAR) CONNECTIVE TISSUE



Diagram 6. Dense Irregular Connective Tissue

- **Abundant collagen fibers**; few cells
 - Characterized by more fibers, few cells
- Contains mostly collagen fibers
- Cells are sparse and typically of a single type (fibroblasts) as primary cell type

- Tissue contains relatively little ground substance
- Fibers are arranged in bundles oriented in various directions
 - Allows the organ to withstand excessive stretching and distortion
- High collagen proportions - significant tensile strength

Location & Function:

- Skin contains a relatively thick layer of connective tissue called stratum reticulare of the dermis
 - Provides resistance to tearing from multidirectional stretching forces
- Hollow organs (like the intestinal track) also possess a distinct layer of this tissue called the **submucosa**
- The fiber bundles course through in varying planes found in organ capsules and periosteum

NOTE!

It is called **irregular** due to the fibers arranged in various directions.

C. DENSE (REGULAR) CONNECTIVE TISSUE

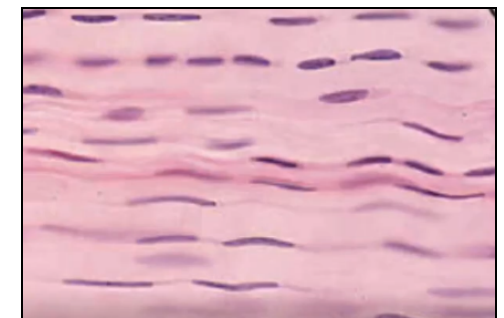


Diagram 7. Dense Regular Connective Tissue

- Characterized by ordered and densely packed arrays of fibers and cells Little ground substance
- Main functional components of tendons, ligaments, and aponeuroses
- **Fibers arranged in parallel array** - densely packed for maximum strength
- Fibroblasts (tenocytes) packed between fiber bundles
- Fibers are the prominent structural feature Provides high tensile strength in a single direction

VII. TENDONS

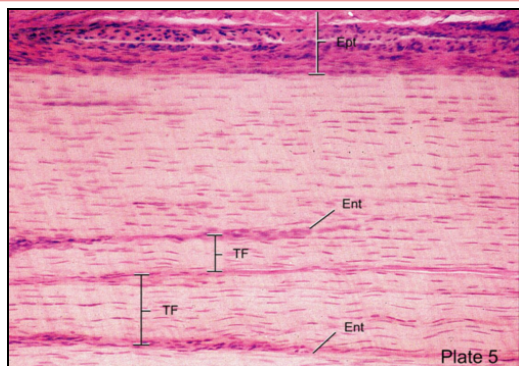


Diagram 8. Tendon (Epitendineum & Endotendineum)

- Cord-like structures attaching muscle to bone
- Parallel bundles of collagen fibers arranged in rows with tenocytes (fibroblasts) in between
- Collagen fibrils appear eosinophilic (pink) in H&E-stained sections
 - Depending on the plane of section, tenocytes appear stellate
- In TEM sections, they are parallel to the long axis of the tendons
- **Epitendineum:**
 - specialized ECM that surrounds the tendon, separating tendon from

loose/thin connective tissue capsule

- Collagen fibers don't appear orderly
- **Endotendineum:**
 - divide tendon into fascicles
 - Contains small blood vessels and nerves

VIII. LIGAMENTS

- Less regularly arranged than tendons.
- Fibers and fibroblasts arranged in parallel; join bone to bone.
- Some require elasticity (e.g. spinal column ligaments).
- Although collagen is the major fiber, some ligaments associated with the spinal column, like the spinal ligaments may contain more elastic fibers; fewer collagen fibers, which are called **elastic ligaments**.

IX. APONEUROSSES

- Broad, flattened tendons.
- Connect bone to muscle or other structures to keep it in place.
- Fibers in parallel arrays arranged in multiple layers.
- Fibers are arranged in regular arrays thus making it a dense regular connective tissue.
- Bundles in one layer oriented at 90 degrees to neighboring layers.
- Orthogonal array also found in cornea (responsible for transparency).

NOTE!

Tendons (*Ept* - Epitendineum, *Ent* - Endotendineum, *TF* - Tendon Fibers)
Ligaments (Horizontal Blue Part), and
Aponeuroses

X. CONNECTIVE TISSUE FIBERS

- Present in varying amount and depending on the structural needs or function of the tissue
- Each fiber is produced by the fibroblast and is composed of proteins consisting of peptide chains
- **Collagen fibers** are the **most abundant type** of these connective tissues
- **Three Principal Types of Connective Tissue Fibers:**
 - **Collagen fibers** - most abundant; provide tensile strength and flexibility
 - **Reticular fibers** - fine branching network; support cellular scaffolding
 - **Elastic fibers** - allow stretch and recoil; interwoven with collagen

A. COLLAGEN FIBERS

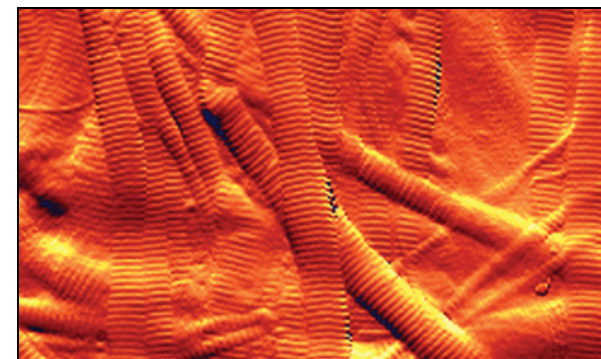


Diagram 9. Collagen Fibers

- **Most abundant structural component of connective tissue**
- Provide flexibility and remarkable high-tensile strength
- **Light Microscopy:**
 - wavy structures of variable width and indeterminate length
- **Electron Microscopy:**

- Bundles of fine thread subunits called collagen fibrils with a characteristic 68-nm banding pattern and are relatively uniform in diameter
- The 68-nm banding pattern can be observed when fibers are stained with osmium or other heavy metals examined with TEM

HISTOLOGICAL IDENTIFICATION

- **H&E Stain:**
 - Stain with **eosin** → appear pink/eosinophilic
- **Mallory trichrome:**
 - Stain with **aniline blue** → appear blue
- **Masson trichrome:**
 - stain with **light green** → appear green

MAJOR COLLAGEN TYPES

Type	LOCATION	FUNCTION
I	Skin, Bone, Tendon, Ligaments, Dentin, Organ Capsules	• Resistance to force, tension, stretch; • Also the most common collagen type (90%).
II	Cartilage (Hyaline & Elastic), Notochord, Intervertebral Disc	• Resistance to pressure
III	Loose CT, Organs (Uterus, Liver, Kidney, Spleen),	• Forms reticular fibers; Provides supportive

	Smooth Muscle, Blood Vessels, Fetal Skin, Lymphoid Tissue	scaffolding for specialized cells of various organs and blood vessels
IV	All Basal Laminae of Epithelial Cells; GI Tract, Kidney, Glomeruli, Lens Capsule, Muscle Cells	• Support and filtration barrier; Contributes to the filtration barrier in kidneys; Helps in attachment of connective tissues
VII	Anchoring Fibrils; Skin, Eye, Uterus, Esophagus	• Secures basal lamina to connective tissue fibers.

NOTE!

All fibers are produced by **fibroblasts**.

B. RETICULAR FIBERS

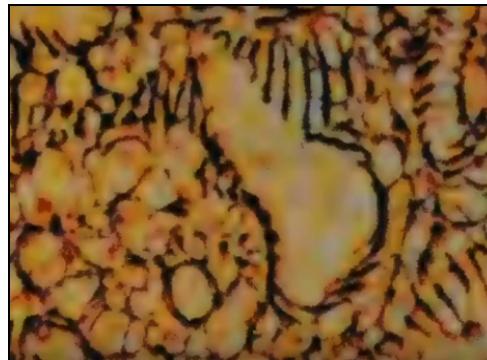


Diagram 10. Reticular Fibers

- Composed of **Type III Collagen**
- Provide supporting framework for cells in tissues and organs
- Individual fibrils: 68-nm banding pattern (same as Type I Collagen)
- Narrower diameter (~20 nm); exhibiting branching pattern
- Typically do not bundle into thick fibers

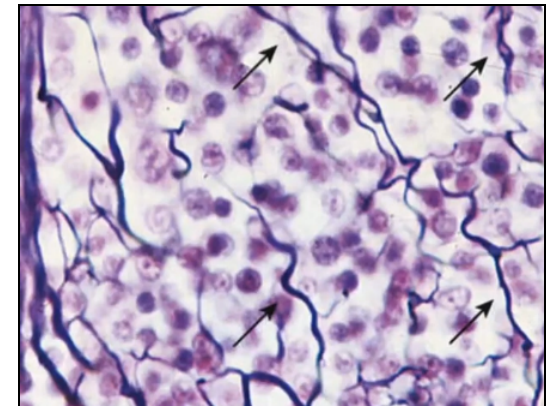


Diagram 11. Reticular Fibers in a Lymph Node

DISTRIBUTION & ARRANGEMENT

- Named for their **mesh-like network pattern**
- In loose connective tissue, the network of reticular fibers are found at the **boundary of connective tissue and epithelium**

SUPPORT FUNCTIONS

- Support **epithelial cells, adipocytes, and muscle cells**
- Function as supporting stroma in hemopoietic and lymphatic tissues, but NOT in the thymus

DEVELOPMENTAL ROLE

- Found in **embryonic tissues**
 - The prevalence of reticular fibers is an indicator of **tissue immaturity**

- Prominent in **initial stages of wound healing** and **scar tissue formation**
 - Provide some mechanical strength (like **scaffolding**)
- As development and wound healing progress, they are replaced by stronger **Type I Collagen fibers**

PRODUCING CELLS

- Reticular cells** (supporting stroma in lymphatic/hemopoietic tissues)
- In other tissues, it is produced by **fibroblasts**, **smooth muscle cells**, and **hepatocytes**

HISTOLOGICAL IDENTIFICATION

- H&E Stain:**
 - Cannot be identified (appear similar to collagen)
 - Needs **special stains**, which give a **thin thread-like appearance**
- Periodic acid-Schiff Stain:**
 - Stain magenta/purple (PAS-positive)
- Silver stains (Gomori, Wilder):**
 - Appear black = argyrophilic

C. ELASTIC FIBERS

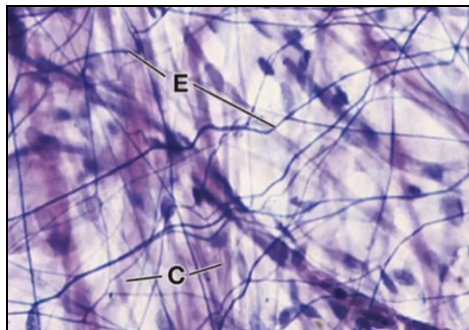


Diagram 12. Collagen (C) and Elastic (E) Fibers

- Allow tissues to **respond to stretch and distension**
- Thinner than collagen fibers**; arranged in branching pattern forming a 3D network
- Elastin is **~1,000x more flexible** than collagen fibers Interwoven with collagen fibers to limit distensibility and prevent tearing
- Produced by:**
 - Fibroblasts
 - Smooth muscle cells
 - Endothelial cells
 - Chondrocytes
- Located in the:**
 - Large arteries
 - True vocal cords
 - Ligamentum nuchae
 - Ligamentum flava

HISTOLOGICAL IDENTIFICATION

- H&E Stain:**
 - stains with eosin but not well
 - may appear refractile with certain fixatives
- Special stains:**
 - orcin or resorcin-fuchsin selectively stain elastic fibers

XI. CLINICAL CORRELATION: COLLAGENOPATHIES

- Collagen diseases caused by genetic mutations affecting collagen gene expression.

Type Being Affected	DISEASE	EFFECTS
I	Osteogenesis Imperfecta	Brittle bones, Frequent fractures

II	Kniest dysplasia	Short stature, Joint mobility issues, Blindness
III	Ehlers-Danlos syndrome	Hypermobile joints, Dislocations, Fragile skin
IV	Alport syndrome	Kidney disease, Hearing loss, Ocular lesions
VII	Kindler syndrome	Skin blistering, Scarring

XII. EXTRACELLULAR MATRIX

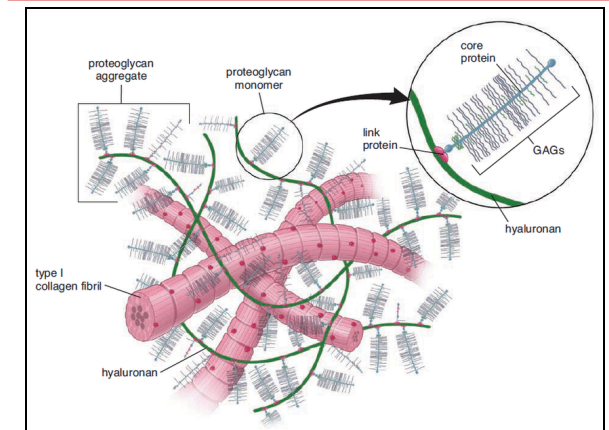


Diagram 14. Extracellular Matrix Components

- Structural network surrounding and supporting cell in connective tissue
- Two Main Components:**
 - Fibers**
 - Collagen
 - Elastic
 - Reticular
 - Ground Substance**
 - GAGs

- Proteoglycans
- Glycoproteins

Functions

- **Mechanical support and tensile strength**
- Allows **cell migration and communication**
- Regulates **cell differentiation and development**

XIII. GROUND SUBSTANCE

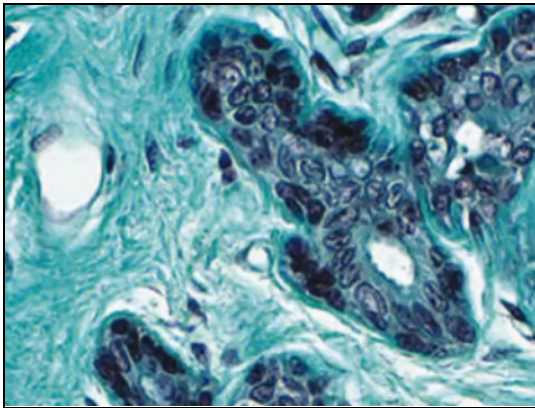


Diagram 13. Ground Substance w/ Alcian Blue Stain

- Fills spaces between cells and fibers.
- **Composition:**
 - Glycosaminoglycans (GAGs)
 - Proteoglycans
 - Multiadhesive glycoprotein
- **Properties:**
 - Viscous
 - Gel-like
 - High water content

HISTOLOGICAL IDENTIFICATION

- H&E Stain:** Lost during processing - appears as empty space.
- PAS Stain:** Magenta/Purple
- Alcian blue:** Stains blue

- Toluidine blue:** Metachromasia (purple instead of blue)

E. GLYCOSAMINOGLYCANS (GAGs)

- Long chain polysaccharides that are highly **negatively charged** → attracts water → form gel
- The **gel** provides **cushioning** and allow **diffusion of materials** throughout connective tissue

IMPORTANT GAGs	
GAG	LOCATION
Hyaluronan / Hyaluronic Acid	• Most abundant • Synovial fluid, cartilage
Chondroitin sulfate	• Cartilage, bone
Heparan sulfate	• Basement membranes
Dermatan sulfate	• Skin, blood vessels
Keratan sulfate	• Cartilage, cornea

B. PROTEOGLYCANS

- Glycosaminoglycans attached to a core protein.
- Form large molecules that trap water.

PROTEOGLYCANS	DESCRIPTION
Aggrecan	Most Important; Found in cartilage; provides cushioning

Decorin	Organizes collagen fibers
Syndecan	Anchors cells to ECM

C. MULTIADHESIVE GLYCOPROTEINS

- Link cells to **ECM**.

MULTIADHESIVE GLYCOPROTEINS	DESCRIPTION
Fibronectin	Helps cells attach to ECM
Laminin	Found in basement membranes; anchors epithelial
Tenascin	Important in embryonic development and wound healing

XIV. CONNECTIVE TISSUE CELLS

Resident Cells (Permanent)

- Fibroblasts
- Adipocytes
- Macrophages
- Mast cells
- Adult stem cells
 - Permanent residents of the tissue.

Wandering Cells (From Blood):

- Lymphocytes
- Plasma cells
- Neutrophils
- Eosinophils
- Basophils

- Monocytes
- Migrate from the blood in response to injury or inflammation.

A. FIBROBLASTS

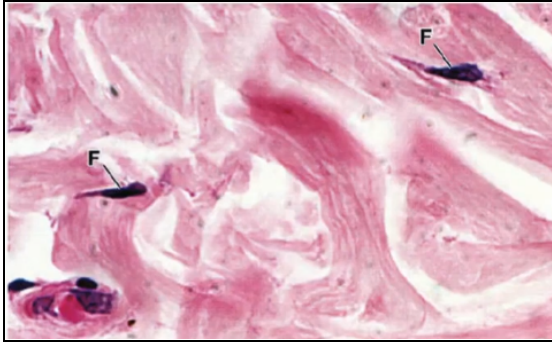


Diagram 15. Fibroblasts in Connective Tissue

- Principal cell of connective tissue.
- Synthesizes all ECM components: collagen, elastic fibers, reticular fibers, and ground substance.

Myofibroblasts

- Modified fibroblasts with contractile properties.
- Important in wound contraction during tissue repair.

HISTOLOGICAL IDENTIFICATION (H&E)

- Only nucleus visible - elongated or spindle-shaped.
- Cytoplasm not visible (blends with surrounding collagen).
- During wound repair: cytoplasm more visible and basophilic.
- Since fibroblasts are more active, causing an increase in rough endoplasmic reticulum.

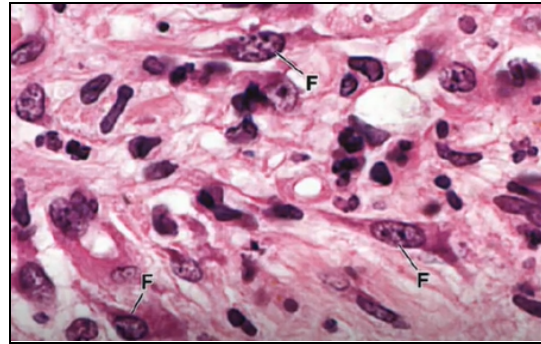


Diagram 16. Fibroblast during Wound Repair

B. ADIPOCYTES

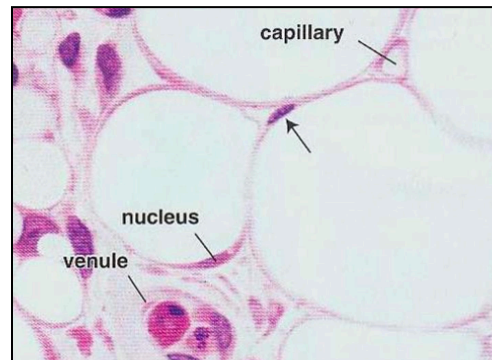


Diagram 17. Adipocytes

- Specialized connective tissue cells for lipid storage
- Differentiate from mesenchymal stem cells
- Located through loose connective tissue as individual cells or groups forming adipose tissue

HISTOLOGICAL IDENTIFICATION (H&E)

- Appear as large, empty circular spaces
 - Lipid dissolved after tissue processing
- Thin rim of cytoplasm a periphery
- Flattened nucleus pushed to one side
 - Classing "signet-ring" appearance

C. MACROPHAGES

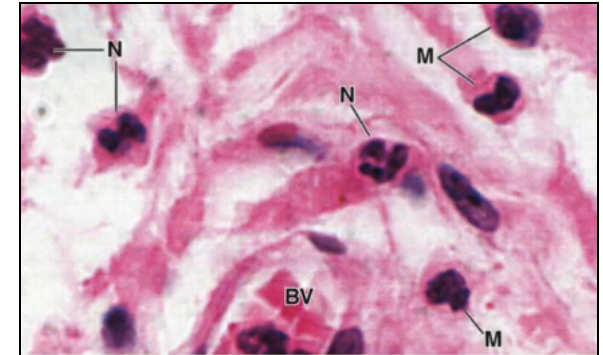


Diagram 18. Macrophages

- Phagocytic cells derived from blood monocytes
 - Mature form of monocytes
- Defense, cleanup, and antigen presentation
- First responders to tissue injury after the neutrophil

HISTOLOGICAL IDENTIFICATION (H&E)

- Kidney-shaped or indented nucleus
- More cytoplasm than fibroblasts
- Difficult to distinguish with fibroblasts in routine preparations

D. MAST CELLS

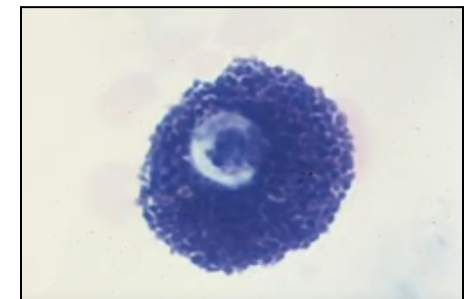


Diagram 19. Mast Cell

- Large ovoid cells (20–30 micrometers) filled with granules containing histamine and heparin
- Mediate allergic acid and inflammatory response
- Are related to basophils
- Located in skin, mucous membranes, and around blood vessels

HISTOLOGICAL IDENTIFICATION

- **H&E:** granules often lost during processing
- **Toluidine blue:** granules stain purple (metachromasia)

XV. MONONUCLEAR PHAGOCYTE SYSTEM

- A system of phagocytic cells derived from monocytes in the bone marrow.
- It is now accepted that the cells from the MPS arise from two distinct populations of progenitor cells in separate waves during hemopoiesis.

FIRST POPULATION

- The early population derives from the erythro-myeloid progenitor cells in the yolk sac and primitive hemopoietic cavity
 - First seen at **2 weeks of gestation** and develop independently from blood monocytes
 - They migrate into developing tissues and organs and become **resident macrophages**

SECOND POPULATION

- The second population originates from the bone marrow hematopoietic stem cells which give rise to definitive hematopoietic macrophages
 - The monocyte first appears in the circulation much later in

development, about at the 17th week of gestation

- Basically a network of immune cells that serves as the body's first line of defense by identifying and destroying foreign substances, pathogen, and damaged cells

Cells of the Mononuclear Phagocyte System	
Cells	Location
Monocytes and their precursors in bone marrow: monoblasts and promonocytes	Blood and bone marrow
Resident macrophages	Connective tissue, spleen, lymph nodes, bone marrow, adipose tissue, and thymus
Dendritic cells	Lymph nodes, spleen
Stellate sinusoidal macrophages (Kupffer cell)	Liver
Alveolar macrophages	Lungs
Mesangial cells	Kidney
Placental macrophages (Hofbauer cell)	Placenta
Pleural and peritoneal macrophages	Serous cavities
Microglia	Central nervous system
Langerhans cells	Epidermis of skin, oral mucosa, foreskin, female genital epithelium
Intestinal macrophages	Gastrointestinal tract: submucosa and lamina propria
Peritubular and interstitial macrophages	Testis
Osteoclasts (originate from hemopoietic progenitor cells)	Bone
Fibroblast-derived macrophages (originate from mesenchymal cells)	Lamina propria of intestine, endometrium of uterus
Multinucleated giant cells (e.g., foreign-body giant cells, Langhans giant cells; originate from fusion of several macrophages)	Pathologic granulomas: suture granuloma, tuberculosis

NOTE!

- Depending on the different locations of the bodies, they have different names and cell types.
- But essentially, they are still part of the mononuclear phagocyte system

A. ADULT STEM CELLS

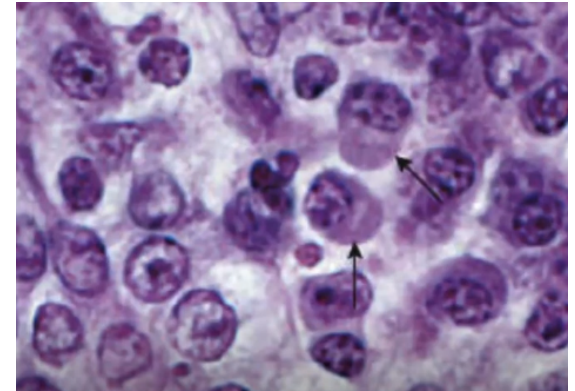


Diagram 20. Plasma Cell

- Found in bone marrow and various tissues/organs.
- Differentiate into specialized connective tissue cells.
- Important in tissue repair and regeneration.

B. PERICYTES (PERIVASCULAR CELLS)

- Wrap around capillaries and venules.
- Act as mesenchymal stem cells.
- Differentiate into various cell types during wound healing and new blood vessel development.

C. WANDERING CELLS

- Migrate to the blood from the connective tissue in response to injury or inflammation, which includes:

CELL	FUNCTION/DESCRIPTION
Lymphocytes	Immune response; Smallest wandering cells

Plasma Cells	Produce antibodies; Derived from B lymphocytes; "Clock face" chromatin and basophilic cytoplasm (due to extensive rER)
Neutrophils, Basophils, Eosinophil	Migrate to sites of inflammation
Monocytes	Differentiate into macrophages in tissues
Plasma Cell	"Clock Face" Chromatin

XVI. ADIPOSE TISSUE

- Specialized type of connective tissue for energy storage.
- Three types: **White, Brown and Beige**
- Functions:**
 - Energy storage
 - Insulation
 - Cushioning
 - Hormone Production

A. WHITE ADIPOSE TISSUE

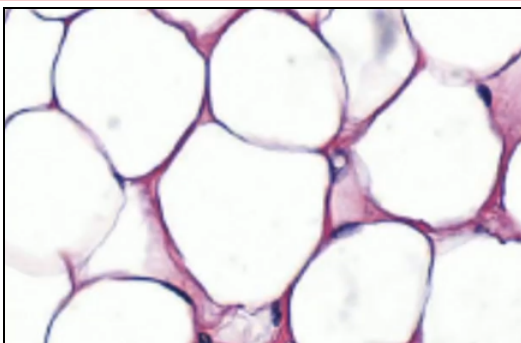


Diagram 21. White Adipose Tissue

- Most common, making up 10% of the body weight.

- Has large unilocular adipocytes with one lipid per cell.
- Functions:**
 - Energy storage (primary fuel reserve)
 - Hormone secretion
 - Ex. **Leptin** helps regulate appetite and homeostasis.
- Thermal insulation
- Mechanical cushioning of organs

DISTRIBUTION

- Subcutaneous tissue (hypodermis).
- Surrounding visceral organs (omentum, mesentery).
- Retroperitoneal and perinephric regions.

HISTOLOGICAL FEATURES (H&E)

- Unilocular cells - single large lipid droplet per cell.
- Lipid dissolved during processing → large empty spaces.
- Thin rim of cytoplasm at periphery.
- Flattened nucleus displaced to periphery (signet-ring appearance).

B. BROWN ADIPOSE TISSUE

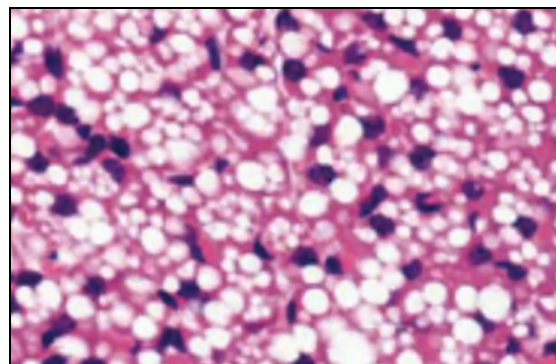


Diagram 22. Brown Adipose Tissue

- Has **multilipid droplets** per cell

- Main function is **heat production through thermogenesis**
 - Non-shivering thermogenesis
 - Mitochondria uncouple oxidative phosphorylation via UCP-1 (thermogenin)
 - Energy is released as heat** rather than ATP

LOCATION

- Abundant in newborn/infants** (critical for thermoregulation):
 - Neck
 - Back
 - Shoulders
- Persists in adults:**
 - Neck
 - Supraclavicular region
 - Aorta (axilla)
 - Kidney (perirenal region)
- Highly vascularized** → characteristic **brown color**

C. BEIGE ADIPOSE TISSUE



Diagram 23. Beige Adipose Tissue

- Intermediate between white and brown adipose tissue.
- Also called **brite (brown-in-white) adipose tissue**.

- **Can transform from white to brown-like tissue in response to:**
 - Cold exposure
 - Exercise
 - Hormones (norepinephrine)
- **Function:**
 - **Adaptive thermogenesis**
 - Heat production when needed.
 - **Clinical Relevance:**
 - Potential target for obesity and metabolic disease treatment.

XVII. REFERENCES

- Pawlina, W. (2026) Histology: A Text and Atlas with Correlated Cell and Molecular Biology (9th ed). Walters Kluwer